

Exploring Alpha Generation through Fixed Income Arbitrages

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Abstract

This study investigates whether textbook arbitrage strategies in fixed income markets generate statistically significant alpha after controlling for systematic risk. It explores the risk factors driving their returns by integrating established asset pricing frameworks with machine learning techniques for factor selection. Time-varying exposures are estimated using a state-space model with observable regime shifts, and model plausibility is assessed via Bayesian asset pricing tests. The project also analyzes cross-strategy linkages to detect common risk exposures and frictions, contributing to the understanding of persistent mispricings and limits to arbitrage.

I. Introduction

Arbitrage strategies in fixed income markets represent a powerful lens through which to study market inefficiencies, funding frictions, and liquidity constraints. While equity markets have been extensively studied through the lens of factor models and asset pricing anomalies—leading to what has been called a “factor zoo”—fixed income arbitrage remains comparatively underexplored, especially in terms of understanding which economic forces drive their returns, and whether they yield abnormal profits beyond compensation for systematic risk.

To the best of our knowledge, this project provides the first unified empirical study of four textbook arbitrage strategies in fixed income, each with deterministic cash flows and clearly defined convergence mechanisms: (i) the CDS-bond basis, (ii) the Italian Inflation-Linked bond basis, (iii) the CDS index Skew (CDS Index vs. single-name constituents), and (iv) 0-100% Index Tranches vs. single-names constituents arbitrage. These trades represent arbitrages in the strict textbook sense: regardless of default or credit events, the net payoff is riskless once executed correctly.

Despite their structural appeal, these arbitrages have frequently exhibited persistent pricing gaps, even during non-crisis periods. Notably, the inflation-linked bond basis between BTP€i or BTP Italia and nominal BTPs has remained significant for over a decade, in contrast with the U.S. TIPS market, where the basis studied by Fleckenstein et al. (2014) has nearly disappeared. To our knowledge, this is the first academic study to systematically analyze such arbitrages in the Italian inflation-linked market, introducing novel evidence on segmentation, credit risk premia, and the pricing of embedded features such as inflation floors.

These deviations raise fundamental questions about market efficiency, the ability of arbitrageurs to eliminate mispricings, and the role of constraints such as balance sheet costs, market segmentation, and funding liquidity. Yet, the existing literature has generally treated

these arbitrages in isolation, has rarely assessed them through formal factor models, and has offered limited insights into their interaction or shared risk drivers.

This study aims to answer the following research questions:

1. Which risk factors systematically explain the returns of the four arbitrage strategies described above?
2. Do these arbitrage strategies produce statistically and economically significant alpha, after adjusting for relevant sources of systematic risk?
3. Are there interdependencies across arbitrage strategies that indicate capital reallocation or slow-moving arbitrage capital, as suggested by Duffie (2010)?

To answer these questions, the study combines high-frequency bond and CDS data from European markets with modern econometric tools tailored to the challenges of high-dimensionality, multicollinearity, and time-varying exposure. First, it integrates insights from the asset pricing and hedge fund literature with modern econometric and machine learning techniques. In particular, we employ Elastic Net regularization to select the most relevant predictors from a large pool of macro-financial and market variables, addressing the limitations of previous studies that relied on fixed sets of explanatory variables across all strategies. Second, we conduct formal model comparisons using the Bayesian asset pricing test of Barillas and Shanken (2018). Third, we adopt a state-space modeling framework with Kalman filtering and observable regime shifts, allowing for time-varying risk exposures and better identification of volatility-driven or liquidity-driven beta instability. Fourth, we perform robustness checks based on block bootstrap and rolling-window inference. These approaches collectively enhance the credibility of our inference in finite samples and under model uncertainty.

Finally, by jointly analyzing the four arbitrages, the project will evaluate cross-strategy interdependence, investigating whether these mispricings share common risk components or

reflect constraints related to capital mobility and funding—an open question in light of the slow-moving capital hypothesis proposed by Duffie (2010).

By combining institutional detail, theoretical precision, and advanced empirical methods, the project seeks to clarify when and why fixed income arbitrages persist, whether they are systematically priced, and what they reveal about the limits to arbitrage in modern financial markets.

II. Literature Review

Duarte et al. (2007) conduct an empirical investigation of fixed-income arbitrage strategies and to assess whether such strategies deliver alpha or merely reflect “picking up nickels in front of a steamroller.” They studied six strategies: swap spread, yield curve, mortgage, volatility, and capital structure arbitrages. After risk-adjusting for both equity and bond market factors, they found that arbitrage strategies on the yield curve, mortgage, and capital structure produced statistically significant alpha. While the paper has the undeniable merit of being among the first to conduct such an analysis, none of the strategies studied can be considered true arbitrages: they are in fact trading strategies that have historically yielded positive average returns but, even if held to maturity, they don’t guarantee a positive return. By contrast, the four arbitrage strategies analyzed in this project are textbook arbitrages. Another important limitation concerns the selection of risk factors which—despite including a wide set—remains arbitrary, and the fact that exposures are modeled with static betas.

The inflation-linked bond arbitrage was instead analyzed by Fleckenstein et al. (2014) who identified a systematic mispricing between TIPS and nominal Treasuries. The authors, analyzing data from 2004 to 2009, identified a mispricing between TIPS and Treasuries averaging 60bps, with peaks of 200bps on some maturities during the financial crisis. Their results showed that the basis was negatively correlated with the supply of both bonds and total global assets held by hedge funds, while positively correlated with “Repo fails” risk. Credit risk and lower liquidity of TIPS (measured as the ratio of daily traded volumes

between the two bonds) were not statistically significant. Today, however, this basis has effectively vanished—compressed to just a few basis points. Consistently with the literature (Duffie, 2010; Brunnermeier and Pedersen, 2009), we believe that much of this is due to the massive liquidity injections during years of quantitative easing, as well as the growing number of institutions (beyond hedge funds) engaging in the arbitrage.

In contrast, this arbitrage still persists in the Italian inflation-linked market (BTP€i and BTP Italia). It cannot be explained either by repo rates (which are the same as for nominal BTPs, since both are accepted as general collateral), nor by credit risk factors such as ballooning effects (which are absent in BTP Italia due to the semiannual revaluation of principal). To the best of our knowledge, this is the first academic study to investigate the inflation-linked basis in the Italian government bond market.

The CDS-bond basis has also been extensively studied. Hull et al. (2004) and Longstaff et al. (2005) document the typical small positive basis in the pre-crisis period, while Mitchell and Pulvino (2012) show that during 2007–2009, tightening financial constraints led to persistent large negative bases. These papers highlight the role of funding frictions and mark-to-market losses under limited arbitrage. However, more recent work such as Bai and Collin-Dufresne (2019) and Fontaine and Nolin (2019) shows that even in normal market conditions, certain categories of bonds (e.g., illiquid or non-ECB eligible) still exhibit persistent basis violations. This motivates a cross-sectional study of basis behavior across liquidity and eligibility buckets, as proposed in this project.

The third arbitrage arises when the spread of a CDS index differs from the duration-weighted average spread of its single-name CDS constituents. As highlighted by Junge and Trolle (2015) and Boyarchenko et al. (2016), this basis tends to arise due to market segmentation and the illiquidity of certain single names. However, neither study formalizes an asset pricing model to explain this arbitrage, nor do they test whether the observed spread differentials can be attributed to priced risk factors or result in statistically significant alpha.

Another key paper is Sadka (2010), which does not analyze individual arbitrage strategies

but rather the performance of hedge funds that invest exclusively in fixed-income arbitrage. Sadka uses as risk factors those suggested in the literature by Fung and Hsieh (2002) (including, in addition to the classic Fama-French three factors, trend-following returns across various asset classes), and adds a proprietary liquidity factor derived from high-frequency TAQ data.

Other studies have also attempted to identify risk factors for fixed-income strategies. Notably, Ludvigson and Ng (2009b) and Ludvigson and Ng (2009a) employ principal component analysis (PCA) and dynamic factor models (DFM) to identify latent macroeconomic drivers of bond premia, but the interpretability of PCA components remains a challenge.

To avoid arbitrary factor selection, we follow the recommendation of Giglio et al. (2022) to combine machine learning with factor models, addressing issues of overfitting and multicollinearity in high-dimensional settings. We use the Elastic Net estimator (Zou and Hastie, 2005) to select a sparse set of predictors from a large pool of market-based and macroeconomic variables. Elastic Net is preferred over LASSO due to its superior performance when predictors are highly correlated, as is typical in fixed income.

III. Research Methodology

We begin by constructing return indices for each of the four arbitrage strategies, following the dynamic rolling fund approach of Duarte et al. (2007), where each trade is modeled as a separate fund. Monthly strategy returns are computed as the equal-weighted average of open positions that are initiated whenever the basis exceed a predefined threshold. The positions are closed upon basis convergence or at maturity. For the first arbitrage, we include all Italian inflation-linked securities issued between January 2015 and May 2025. We use zero-coupon inflation swap curves to construct the corresponding inflation asset swap to convert bond's cash flows into fixed. The resulting yield is compared to the yield of the closest maturity nominal BTP to calculate the basis. Data are sourced from Bloomberg: Official MOT (Mercato Obbligazionario Telematico) prices are used for bond quotations, while ICPL

(Interdealer Composite Pricing Level) quotes are used for inflation swaps.

Regarding the CDS-bond basis arbitrage, we include for liquidity reasons, only issuers whose single-name CDS contracts are constituents of the European iTraxx or U.S. CDX Indices. Each bond is matched with its corresponding CDS of similar maturity and seniority. Bonds are grouped into distinct liquidity buckets. For the Index Skew and 0–100% Tranches versus single names CDS arbitrages, we use both on-the-run and one series off-the-run iTraxx and CDX indices for liquidity purposes. For all CDS instruments, we use settlement prices provided by ICE (Intercontinental Exchange).

Once historical return series have been constructed for all four strategies, we proceed to address our first research question: *What risk factors drive the returns of these arbitrage strategies?* We begin the empirical analysis by testing whether standard risk factors identified in the asset pricing literature explain the returns of the arbitrage strategies. The first specification draws from Duarte et al. (2007) includes the following factors:

$$R_{it} = \alpha + \beta_1 RM_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 UMD_t + \beta_5 RS_t + \beta_6 R2_t + \beta_7 R5_t + \beta_8 R10_t + \beta_9 RI_t + \beta_{10} RB_t + \varepsilon_t,$$

where RM_t is the excess return on the CRSP value-weighted portfolio, and SMB_t , HML_t , and UMD_t are the Fama-French size, value, and momentum factors, respectively. RS_t denotes the excess return on an S&P index of bank stocks. $R2_t$, $R5_t$, and $R10_t$ are the returns on the CRSP Fama Treasury bond portfolios for two-year, five-year, and ten-year maturities. RI_t and RB_t are the excess returns on the Merrill Lynch indices for A/BAA-rated industrial and bank bonds, respectively.

Subsequently, we test the factors proposed by Fung and Hsieh (2002) and Sadka (2010), widely used in the hedge fund literature. Specifically, we estimate the following regression:

$$R_{it} = \alpha + \beta_1 (MKT-RF)_t + \beta_2 SMB_t + \beta_3 \Delta TERM_t + \beta_4 \Delta CREDIT_t + \beta_5 PTF SBD_t + \beta_6 PTF SFX_t + \beta_7 PTF SCOM_t + \beta_8 Liquidity_t + \varepsilon_t,$$

where $(MKT-RF)_t$ is the market excess return, $\Delta TERM_t$ and $\Delta CREDIT_t$ are the changes in the term and credit spread, respectively. $PTF SBD_t$, $PTF SFX_t$, and $PTF SCOM_t$

are the trend-following returns on bond, currency, and commodity futures, respectively. $Liquidity_t$ is the liquidity innovation factor extracted from high-frequency TAQ data as proposed by Sadka (2010).

These two initial OLS regressions serve as a diagnostic tool to test whether commonly cited risk factors from the literature significantly explain the returns of each arbitrage strategy. In particular, we assess the explanatory power of the benchmark models and evaluate whether the intercept (α) is statistically different from zero, thereby indicating the presence of unexplained abnormal performance.

Beyond these pre-identified factors, we aim to investigate whether alternative or complementary risk drivers better capture the economic mechanisms underlying these strategies. To this end, we collect time-series data on a broad set of candidate variables, including sovereign yield spreads (e.g., BTP/Bund 10-year spread, Schatz-Bobl-Bund ASW spreads), equity market excess returns (e.g., EuroStoxx50, S&P 500, EuroStoxx Banks), liquidity proxies (e.g., bid-ask spreads, bond amount outstanding), inflation expectations (e.g., breakeven inflation, 5y5y inflation swaps), credit risk indicators (e.g., iTraxx Main and Xover 5Y/3Y flatteners, Main-SnrFin basis), and market volatility (e.g., VIX index).

To select the most relevant predictors among these 30–40 variables, we apply the Elastic Net regression framework, a regularized machine learning approach widely used in empirical finance. Elastic Net combines the sparsity-inducing properties of the LASSO (Least Absolute Shrinkage and Selection Operator) with the shrinkage of Ridge regression. It is particularly suited to our setting due to its ability to handle multicollinearity among predictors and to retain groups of correlated regressors—features that LASSO alone may fail to address (Zou and Hastie (2005)). For each strategy, the Elastic Net selects a parsimonious set of approximately five to six variables that exhibit the strongest predictive relationships with monthly returns over the full sample period.

Following the Elastic Net selection, we formally assess whether the resulting factor structure provides superior pricing performance compared to standard benchmark models. We

apply the Bayesian model comparison framework of Barillas and Shanken (2018), which evaluates the relative plausibility of two linear factor models under the maintained assumption of correct pricing (i.e., $\alpha = 0$). The posterior odds—summarized by a Bayes Factor—quantify which model better rationalizes observed returns, making the test particularly informative in small samples. For this analysis, we estimate a static linear model via OLS and compute the squared Sharpe ratio of the alpha vector, $SR^2 = \alpha' \Sigma^{-1} \alpha$, where Σ is the residual covariance matrix. A higher Sharpe ratio indicates mispricing, and the Bayesian framework evaluates whether this is more consistent with the benchmark or the extended model.

Once the most relevant factors have been selected for each strategy, we estimate the following time-varying beta model:

$$r_{it} = \alpha_t + \sum_{j=1}^k \beta_{j,t} x_{j,t} + \varepsilon_t,$$

where the exposure $\beta_{j,t}$ is allowed to vary over time and we model it using a state-space representation and estimate it via the Kalman filter. To enhance the model’s responsiveness to abrupt regime changes in market conditions—such as those caused by COVID-19, geopolitical crises, or major central bank interventions—we introduce observable state variables (e.g., dummy indicators for regime shifts) into the transition equation:

$$\beta_{j,t} = \beta_{j,t-1} + \delta_j \cdot \text{State}_t + \nu_{j,t}, \quad \nu_{j,t} \sim \mathcal{N}(0, Q),$$

where: State_t is the observable market condition indicator, and δ captures the discrete shift in exposure induced by the regime.

We do not adopt Markov Switching models due to their reliance on latent and endogenously estimated state transitions, which require substantially larger samples to yield stable estimates. As shown in Krolzig and Krolzig (1997) and Hamilton (2010), consistent estimation of regime-switching models typically demands longer time series ($T > 200$), a requirement that exceeds the size of our sample ($T \approx 120$).

After recursively estimating the parameters via the Kalman filter, we test the statistical significance of the intercept (α) for each strategy using a t-test on the average of the residuals. To account for potential autocorrelation and heteroskedasticity, we compute HAC-consistent

(Newey-West) standard errors.

To evaluate the statistical significance of the intercept (α) and the time-varying factor loadings (β_t), we implement a residual-based bootstrap procedure adapted to state-space models (Durbin and Koopman, 2012). Since standard Kalman filter residuals are filtered estimates rather than true structural shocks, we use the one-step-ahead forecast errors (innovation residuals):

$$\tilde{\varepsilon}_t = r_t - \hat{\alpha} - \sum_{j=1}^k \hat{\beta}_{j,t|t-1} x_{j,t},$$

where $\hat{\beta}_{j,t|t-1}$ denotes the predicted exposure at time t based on information through $t-1$. These innovation residuals are resampled using the stationary block bootstrap method of Politis and Romano (1994), which preserves temporal dependence by drawing blocks with geometrically distributed lengths. For each of the 10,000 bootstrap replications, we generate synthetic return series by combining the resampled residuals with the model-fitted values, and then re-estimate the full state-space model using the Kalman filter. This process yields empirical distributions of both $\hat{\alpha}$ and $\hat{\beta}_{j,t}$ under minimal distributional assumptions.

As an additional control, we implement adaptive rolling window regressions based on local return volatility, as proposed by Fama and French (2010). This technique allows us to assess whether alpha estimates are stable over time and whether their magnitude or significance is sensitive to regime changes.

Finally, following Fleckenstein et al. (2014), we examine the cross-strategy correlation structure. To do so, we regress the changes in the return index of each arbitrage strategy on the contemporaneous changes in the other three. The rationale is that constrained arbitrage capital may lead investors to allocate to the most efficient strategy at any given time, inducing dynamic substitution effects. This mechanism implies that some arbitrages may close with a delay relative to others that are more liquid or more profitable, consistent with the slow-moving capital hypothesis.

IV. Expected Results

This project aims to investigate the presence and persistence of arbitrage opportunities across four textbook fixed income strategies, and to assess whether their returns can be fully explained by systematic risk factors or instead reflect exploitable mispricings. While no empirical results have yet been produced, several expected outcomes are grounded in theoretical priors and the structure of the trades.

First, we expect that each arbitrage strategy will be explained by a distinct set of risk drivers. Previous literature often applied a common set of explanatory variables across strategies (e.g. Sadka (2010), Duarte et al. (2007)), potentially overlooking asset-specific mechanisms. In contrast, our use of Elastic Net regression is expected to reveal a heterogeneous structure of predictors, allowing for a parsimonious and tailored factor model per strategy.

Second, one of the central questions in the literature is whether textbook arbitrage strategies in fixed income markets deliver genuine abnormal returns, or whether they simply reflect compensation for hidden risk exposures. By combining machine learning for factor selection with robust econometric techniques—including a Bayesian model comparison framework—we expect to shed new light on this issue. These tools are designed to overcome limitations of prior studies that relied on fixed factor sets and static linear models, enabling a more rigorous test of whether statistically and economically significant alpha can persist after controlling for time-varying risks.

Third, we expect to observe meaningful time variation in factor loadings, particularly in response to changes in volatility, funding constraints, or market segmentation. The state-space framework with observable regime shifts should help identify periods during which arbitrage activity becomes more constrained or sensitive to macro-financial shocks.

Fourth, although the strategies are conceptually distinct, we expect to identify a limited set of common latent risk factors—such as liquidity, volatility, or funding stress—that affect

multiple arbitrages simultaneously. This would be consistent with the slow-moving capital hypothesis of Duffie (2010), and may help explain substitution or contagion patterns across strategies.

The combination of penalized regression, state-space modeling, and robust inference techniques is expected to yield a more accurate and flexible understanding of the underlying return dynamics. These methods aim to disentangle persistent mispricings from risk-based compensation, offering a sharper distinction than what has been achievable with traditional static models. Clarifying this boundary is crucial for advancing asset pricing theory and has practical implications for capital allocation, regulation, and the sustainability of arbitrage strategies under balance sheet constraints.

V. Timeline

The project will span approximately six months. In June–July 2025, I will collect and process data to construct return indices for each arbitrage strategy. In August, I will perform benchmark regressions and select risk factors using Elastic Net. September will be dedicated to estimating time-varying exposures via Kalman filtering and conducting bootstrap inference. In October, I will complete robustness checks (including Bayesian model comparison and correlation analysis across strategies). The final month, November 2025, will be devoted to writing and preparing the research presentation.

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